

Coding Exercise 2

1. Assume that you have a data set with k 1's and $(n - k)$ 0's. For a given value of p , write a code to test whether the data is generated according to $\text{Binomial}(n, p)$. In other words, compute the p -value of this test numerically without using any in-built libraries (including `scipy.stats.binom`), and verify if the value is the same as received using `scipy.stats.binomtest` command.
2. For a given contingency table with the values (a, b, c, d) , write a code to test whether the table was generated randomly or not. In other words, compute the p -value of the Fisher's exact test numerically without using any in-built libraries, and verify if the value is the same as received using `scipy.stats.fisher_exact` command.
3. For a given data set and a given value of B , perform bootstrapping and generate B datasets. Plot a histogram of the kurtosis of each dataset, and also compute the 5% confidence interval. Verify if the confidence interval is close to the values computed using `scipy.stats.bootstrap` command. (You can generate data using the command `scipy.stats.skewnorm(a=1,size=n)`.)
4. Assume that you are given a dataset $X = \{X_1, \dots, X_n\}$.
 - (a) Write a code to find whether the hypothesis that the kurtosis of X belongs to the 95% confidence interval of the kurtosis of a Gaussian random variable. Verify if the p -value is close to that given by `scipy.stats.monte_carlo_test`.
 - (b) Repeat the above procedure to find whether the hypothesis that the kurtosis of X belongs to the 95% confidence interval of the kurtosis of the dataset in Q3.
 - (c) Repeat this to find whether the hypothesis that the kurtosis of the dataset in Q3 belongs to the 95% confidence interval of X . Do you get similar p -values in both the tests?